

Regional Quantitative Problem-Solving Circle

Solution Key 5: Algorithmic Invariants & Exchange Arguments (Weeks 9–10)

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Term: Fall 2025 – Spring 2026

Rigorous Proofs & Structural Solutions

Problem 1 (Interval Scheduling / Adjacent Exchange — IOI / Kleinberg-Tardos Caliber). Prove that the greedy choice of selecting the compatible interval with the earliest finish time (f_i) maximizes total scheduled requests.

Proof. Let $G = \{g_1, g_2, \dots, g_k\}$ be the set of disjoint intervals selected by the greedy earliest-finish-time algorithm, ordered strictly by finish time such that $f(g_1) \leq f(g_2) \leq \dots \leq f(g_k)$. Let $O = \{o_1, o_2, \dots, o_m\}$ be a hypothetical optimal solution of disjoint intervals, also ordered by finish time such that $f(o_1) \leq f(o_2) \leq \dots \leq f(o_m)$. By definition of optimality, $m \geq k$.

We will prove by mathematical induction that for all integer indices $r \leq k$, the greedy interval finishes no later than the optimal interval: $f(g_r) \leq f(o_r)$.

Base Case ($r = 1$): The greedy algorithm explicitly selects the interval in the entire global set with the absolute minimum finish time. Thus, the finish time of g_1 must be less than or equal to the finish time of any interval in O , including o_1 . Therefore, $f(g_1) \leq f(o_1)$.

Inductive Step: Assume that $f(g_r) \leq f(o_r)$ holds for some integer $r < k$. We must show that $f(g_{r+1}) \leq f(o_{r+1})$. By the definition of a valid schedule, the interval o_{r+1} must start strictly after o_r finishes, yielding $s(o_{r+1}) \geq f(o_r)$. Using our inductive hypothesis $f(g_r) \leq f(o_r)$, we have $s(o_{r+1}) \geq f(g_r)$. This inequality mathematically proves that o_{r+1} is fully compatible with the greedy interval g_r . When the greedy algorithm selected g_{r+1} , it evaluated all intervals compatible with g_r and chose the one minimizing the finish time. Because o_{r+1} was available and compatible, the greedy choice g_{r+1} must finish no later than o_{r+1} . Thus, $f(g_{r+1}) \leq f(o_{r+1})$.

Conclusion via Contradiction: We have established that $f(g_k) \leq f(o_k)$. Now, assume for the sake of contradiction that the optimal schedule is strictly larger than the greedy schedule, meaning $m > k$. This implies there exists at least one additional interval $o_{k+1} \in O$. For O to be valid, o_{k+1} must start after o_k finishes: $s(o_{k+1}) \geq f(o_k)$. Since $f(g_k) \leq f(o_k)$, it strictly follows that $s(o_{k+1}) \geq f(g_k)$. This proves o_{k+1} is compatible with g_k . However, the greedy algorithm only terminates when strictly zero compatible intervals remain. If o_{k+1} existed and was compatible, the greedy algorithm would have selected it, meaning $|G| > k$, which contradicts our definition of G . Since assuming $m > k$ yields a contradiction, and $m \geq k$ by definition of optimality, we must have $m = k$. The greedy algorithm is optimal. $\square$

Problem 2 (Minimizing Maximum Lateness / Inversion Invariant — Algorithmic Tripos). Prove that any schedule with an inversion can be swapped without increasing the maximum lateness $L = \max_i(C_i - d_i)$, proving EDF is globally optimal.

Proof. Consider an optimal schedule O that does not follow the Earliest Deadline First (EDF) order. Therefore, O must contain at least one inversion: an adjacent pair of jobs where job i is scheduled immediately before job j , but their deadlines violate the EDF rule, meaning $d_i > d_j$. Let the processing times be t_i and t_j . Let S be the start time of job i in this schedule. Before the swap, the completion

times and lateness are evaluated as:

$$\begin{aligned} C_i &= S + t_i \\ C_j &= S + t_i + t_j \\ L_i &= C_i - d_i = S + t_i - d_i \\ L_j &= C_j - d_j = S + t_i + t_j - d_j \end{aligned}$$

The maximum lateness localized to this pair is $\max(L_i, L_j)$. Now, apply an exchange perturbation, creating a new schedule O' by swapping job i and job j . Job j now processes first. The new completion times and lateness are:

$$\begin{aligned} C'_j &= S + t_j \\ C'_i &= S + t_j + t_i = C_j \\ L'_j &= C'_j - d_j = S + t_j - d_j \\ L'_i &= C'_i - d_i = S + t_i + t_j - d_i \end{aligned}$$

We must prove that the maximum lateness of the swapped pair does not exceed the original maximum lateness: $\max(L'_i, L'_j) \leq \max(L_i, L_j)$. First, observe L'_i . Since $C'_i = C_j$ and $d_i > d_j$ (the inversion condition), we have:

$$L'_i = C'_i - d_i = C_j - d_i < C_j - d_j = L_j$$

Second, observe L'_j . Because $t_i > 0$, we know $S + t_j < S + t_i + t_j$, giving:

$$L'_j = S + t_j - d_j < S + t_i + t_j - d_j = L_j$$

Because both $L'_i < L_j$ and $L'_j < L_j$, it is strictly guaranteed that $\max(L'_i, L'_j) \leq L_j \leq \max(L_i, L_j)$. Since the completion times of all jobs preceding the pair and all jobs succeeding the pair remain strictly unchanged (as $t_i + t_j = t_j + t_i$), the global maximum lateness L' of the new schedule satisfies $L' \leq L$. By iteratively eliminating all adjacent inversions via swaps, we monotonically transform any schedule into the EDF schedule without ever increasing the maximum lateness. Thus, EDF is optimal. $\square$

Problem 3 (Chip-Firing / Monovariant Termination — Olympiad Caliber). Define $\Phi = \sum_{v=1}^M v^2 \cdot c(v)$. Prove that Φ is a strict monovariant and that the game terminates.

Proof. Let the total potential of the system at any given discrete state be $\Phi = \sum_{v=1}^M v^2 \cdot c(v)$. We evaluate the change in potential, $\Delta\Phi$, after a single legal firing move occurs at an arbitrary internal vertex k . By the rules of the game, a legal firing removes exactly 2 chips from vertex k , adds exactly 1 chip to vertex $k - 1$, and adds exactly 1 chip to vertex $k + 1$. The chip counts at all other vertices remain strictly invariant. The change in the state potential is algebraically isolated to these three vertices:

$$\begin{aligned} \Delta\Phi &= \Phi_{\text{new}} - \Phi_{\text{old}} \\ &= \left((k-1)^2(1) + (k+1)^2(1) - k^2(2) \right) \\ &= \left((k^2 - 2k + 1) + (k^2 + 2k + 1) - 2k^2 \right) \\ &= \left(2k^2 + 2 - 2k^2 \right) \\ &= 2 \end{aligned}$$

This yields a profound structural invariant: every single legal move strictly increases the global potential function by exactly 2, meaning $\Delta\Phi = 2 > 0$. Therefore, Φ is a strictly increasing monovariant. Because

the total number of chips N is strictly conserved and the graph is a finite domain bounded by the maximum vertex label M , the absolute maximum possible potential occurs if all N chips were stacked on the highest indexed vertex M . Thus, $\Phi \leq N \cdot M^2$. Because Φ is an integer sequence strictly increasing by 2 at every step, and is bounded by a finite mathematical ceiling, the sequence of states must terminate by the Extremal Principle. The game ends in a finite number of steps. $\square$